Chapter 13. Identifiable Gaussian Linear SEMs

Graphical models, Lecture 6 (equal error variances, conditional-variance ordering, ANM, the US algorithm)

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1 In General Gaussian SEMs, Directions Are Not Identified

$$\boxed{X \to Y \to Z} \qquad \boxed{X \leftarrow Y \to Z} \qquad \boxed{X \leftarrow Y \leftarrow Z}$$

The three structures can produce the same conditional independences and the same Gaussian distribution (Ch. 9).

General Gaussian SEM: $\epsilon_j \sim N(0, \sigma_j^2)$	Equal-variance Gaussian SEM: $\epsilon_j \sim N(0, \sigma^2)$
if each variable may have its own error variance, several directions are compatible with the same distribution result: CPDAG	if all structural errors are restricted to the same variance, the residual-variance structure differs by direction result: DAG

LiNGAM broke the tie with non-Gaussianity (Ch. 12). This chapter shows that, even for Gaussian data, an assumption on the *error variances* can identify the full DAG.

2 The Equal-Variance Gaussian SEM

Definition: equal-variance Gaussian SEM

Each variable is a linear combination of its parents plus a structural error:

$$X_j = \sum_{k \in \mathrm{pa}(j)} b_{jk} X_k + \epsilon_j, \qquad \epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

$$X = BX + \epsilon, \qquad \Sigma = \sigma^2 (I - B)^{-1} (I - B)^{-\top}.$$

Assumption	Meaning
linearity	effects of the parents are linear combinations
acyclicity	with a suitable ordering, B is strictly lower triangular
independent errors	no hidden common cause among the analyzed variables
common error variance	all structural errors have the same σ^2
nonzero edges	coefficients of the edges in the graph are not 0

Under these assumptions B and the DAG are identified from the observational distribution.

3 Two Variables: The Equal-Variance Assumption Picks the Direction

True direction $X \to Y$:

$$X = \epsilon_X, \qquad Y = bX + \epsilon_Y, \qquad \epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2),$$

so the two structural errors have $\mathrm{Var}(\epsilon_X) = \mathrm{Var}(\epsilon_Y) = \sigma^2$.

Reverse direction $Y \to X$: rewrite the same distribution as $Y = \epsilon'_Y$, $X = cY + \eta$. Then

$$\mathrm{Var}(\epsilon'_Y) = (1 + b^2)\sigma^2, \qquad \mathrm{Var}(\eta) = \frac{\sigma^2}{1 + b^2}.$$

Proof [Supplement]. $\mathrm{Var}Y = b^2\sigma^2 + \sigma^2$. The reverse regression has $c = \mathrm{Cov}(X,Y)/\mathrm{Var}Y = b/(1+b^2)$ and residual variance $\mathrm{Var}(X \mid Y) = \sigma^2 - \dfrac{b^2\sigma^4}{(1+b^2)\sigma^2} = \dfrac{\sigma^2}{1+b^2}$. ■

The two reverse error variances are equal only if $(1+b^2)^2 = 1$, i.e. $b = 0$. **Both directions explain the Gaussian distribution, but only the true direction satisfies the common error variance.**

4 Conditional Variances Reveal the Causal Order

Motivation. If all noise variances are equal, a source variable has the smallest variance, and after conditioning on it the next source has the smallest conditional variance. Repeating this recovers the causal order of a Gaussian SEM.

4.1 The key lemma

Theorem: conditional-variance ordering

In an equal-variance Gaussian SEM, let S be an *ancestral* set (it contains all ancestors of each of its members, e.g. the first few variables of a causal order) and $j \notin S$. Then

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \quad \text{if } \mathrm{pa}(j) \subseteq S, \qquad \mathrm{Var}(X_j \mid X_S) \geq \sigma^2(1 + b_{jk}^2) > \sigma^2 \quad \text{otherwise,}$$

for a suitable parent $k \notin S$ of j.

Proof [Supplement]. Write $X = A\epsilon$ with $A = (I - B)^{-1}$; a_{jl} is the total effect of l on j (sum over directed paths of products of coefficients), $a_{jj} = 1$, and $a_{jl} = 0$ unless l is an ancestor of j or $l = j$. Because S is ancestral, every X_s, $s \in S$, is a function of $(\epsilon_l)_{l \in S}$ only, and conversely $(\epsilon_l)_{l \in S}$ is recovered from X_S (invert the triangular system). Hence, with $X_j = \sum_{l \in S} a_{jl}\epsilon_l + \sum_{l \notin S} a_{jl}\epsilon_l$, the second sum is independent of X_S and the first is determined by it:

$$\mathrm{Var}(X_j \mid X_S) = \sigma^2 \sum_{l \notin S} a_{jl}^2 = \sigma^2\Big(1 + \sum_{l \notin S,\, l \neq j} a_{jl}^2\Big).$$

If $\mathrm{pa}(j) \subseteq S$, every ancestor of j is in S (ancestral), so the sum is empty. Otherwise let k be the parent of j outside S that is *latest* in a causal order. Any directed path from k to j other than the edge $k \to j$ ends with an edge $m \to j$ where m is a parent of j and a descendant of k; such an m is not in S (S is ancestral and $k \notin S$) and is later than k, contradicting the choice of k. So $a_{jk} = b_{jk} \neq 0$, giving the bound. ■

So, given the already ordered variables, a *source* of the remaining graph has the smallest possible conditional variance σ^2, and every non-source has strictly more. This proves:

Theorem: identifiability (Peters and Bühlmann, 2014)

The DAG and B of an equal-variance Gaussian SEM are identified from Σ. A causal order is obtained top-down: $\pi_1 = \arg\min_j \mathrm{Var}(X_j)$, and $\pi_m = \arg\min_{j \notin S} \mathrm{Var}(X_j \mid X_S)$ with $S = \{\pi_1, \ldots, \pi_{m-1}\}$.

4.2 A worked example

Example: three variables

Consider the Gaussian SEM

$$X_1 = \epsilon_1, \qquad X_2 = 0.8X_1 + \epsilon_2, \qquad X_3 = 0.7X_2 + \epsilon_3, \qquad \epsilon_1, \epsilon_2, \epsilon_3 \overset{\text{ind}}{\sim} N(0,1).$$

Step	Variance comparison	Choice
1	$\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = 1.64$, $\mathrm{Var}(X_3) = 1.8036$	X_1
2	$\mathrm{Var}(X_2 \mid X_1) = 1$, $\mathrm{Var}(X_3 \mid X_1) = 1.49$	X_2
3	the last remaining variable	X_3

Here $\mathrm{Var}(X_2) = 0.64 + 1$, $\mathrm{Var}(X_3) = 0.49 \times 1.64 + 1$, and $\mathrm{Var}(X_3 \mid X_1) = 0.49\,\mathrm{Var}(X_2 \mid X_1) + 1 = 1.49$. Result: $X_1 < X_2 < X_3$.

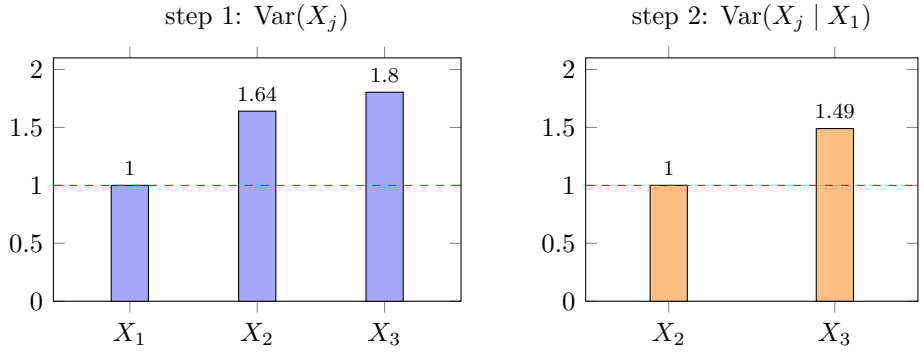


Figure 1: Top-down ordering in the equal-variance example ($\sigma^2 = 1$, red dashed line). At each step the variable whose (conditional) variance equals the common error variance — a source of the remaining graph — is selected: first X_1, then X_2.

4.3 After the order: select parents and edges

Let the estimated order be $X_1 < X_2 < X_3$. Each variable uses only earlier variables as candidate parents:

$$X_2 = \beta_{21}X_1 + \epsilon_2, \qquad X_3 = \beta_{31}X_1 + \beta_{32}X_2 + \epsilon_3.$$

Regression or sparse regression gives $\hat{\beta}_{21} \approx 0.8$, $\hat{\beta}_{31} \approx 0$, $\hat{\beta}_{32} \approx 0.7$, hence

$$X_1 \to X_2 \to X_3.$$

Conditional variances find the order of the variables; regression coefficients select the direct edges within that order.

5 Equal Variance Does Not Mean Equal Observed Variances

Misconception	Correct meaning
all X_j must have the same variance	variation transmitted from the parents makes observed variances differ (as in the example)
standardizing all variables gives equal variance	standardization rescales the structural errors too, so it does not guarantee the equal-variance assumption — it typically destroys it
similar fitted residual variances prove the assumption	only a diagnostic compatible with the chosen DAG; it does not prove the true direction

When the equal-variance assumption is relatively natural: variables of the same kind and unit, whose measurement errors are also of similar size (e.g. expression levels measured on a common scale).

Remark: scale dependence and varsortability

The assumption is not scale-invariant: multiplying one variable by a constant changes its error variance. In many simulated benchmarks marginal variances simply increase along the causal order ("varsortability"), so methods exploiting variances can look excellent there; on real data with arbitrary units this signal may be absent or misleading. Check the variable units and measurement error, and the plausibility of the assumption, before trusting the order.

6 Additive Noise Models and the US Algorithm

6.1 The equal-variance Gaussian SEM is a special ANM

In an additive noise model each variable is

$$X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j, \qquad \epsilon_1, \dots, \epsilon_p \text{ independent.}$$

With S the set of already selected variables, define the expected conditional variance

$$U_j(S) = \mathbb{E}\{\mathrm{Var}(X_j \mid X_S)\}.$$

If S is ancestral and $j \notin S$, then $\epsilon_j \perp\!\!\!\perp X_S$ and

$$U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}].$$

Conditioning set S	Conditional variance
all parents are in S	$U_j(S) = \sigma_j^2$
some parent not yet in S	uncertainty created by the parents is added

6.2 ANMs are identified by an ordering of conditional variances

Let the true causal order be $\pi = (\pi_1, \dots, \pi_p)$ and $S_{m-1} = \{\pi_1, \dots, \pi_{m-1}\}$. If at each step the true next variable has a smaller conditional variance than the other candidates,

$$U_{\pi_m}(S_{m-1}) < U_k(S_{m-1}), \qquad k \notin S_{m-1} \cup \{\pi_m\},$$

then the variables can be identified one by one.

Equal-variance Gaussian SEM	General ANM
$\sigma_1^2 = \cdots = \sigma_p^2$ a variable whose parents are all included has no extra uncertainty, hence the smallest conditional variance (lemma above)	$\sigma_1^2, \dots, \sigma_p^2$ may differ identified as long as the order of conditional variances, combining error variance and parent-induced uncertainty, is maintained

The identifiability of ANMs needs neither linearity nor Gaussianity, but the *correct conditional-variance order*. (For general ANMs with nonlinear f_j there are other identifiability results, based on residual independence as in LiNGAM; the conditional-variance route is the one used here.)

6.3 The US algorithm

Algorithm: US — order by conditional variances, then parents

Step 1: estimate the causal order. Let S be the selected set (initially $\varnothing$).

1. Regress each X_j, $j \notin S$, on X_S (flexibly, if f_j may be nonlinear).
2. Compute the residual variance $\widehat{U}_j(S)$.
3. Select the variable with the smallest value, $\hat{\pi}_m = \arg\min_{j \notin S} \widehat{U}_j(S)$.

Add the selected variable to S and repeat.
Step 2: select the parents. The candidate parents of each variable are the variables before it in the order. For each candidate, test conditional dependence given the other preceding variables:

$$X_{\pi_m} \not\perp\!\!\!\perp X_{\pi_j} \mid X_{\{\pi_1,\ldots,\pi_{m-1}\}\setminus\{\pi_j\}}.$$

If conditional dependence remains, add the edge $\pi_j \to \pi_m$.

US finds the order of directions with conditional variances and selects the actual edges with conditional independence. Within a correct order, the preceding variables other than the parents are non-descendants, so the Markov property guarantees that non-parents are conditionally independent given the rest (and faithfulness that parents are not).

7 Summary

Model	Extra assumption	Clue for the direction
LiNGAM	linear relations with independent non-Gaussian errors	independence of the cause and the regression residual
equal-variance Gaussian SEM	common variance of independent Gaussian errors	order of conditional variances starting from the source
ANM and US	an additive noise model that keeps the conditional-variance order	conditional variances find the order; conditional independence selects the parents

ANM and US allow nonlinear functions and different error variances, but still need acyclicity, no hidden confounding and causal minimality.

Key points.

- A general Gaussian linear SEM cannot distinguish Markov-equivalent directions from observational data.
- If all structural errors have the same variance, the full DAG is identified from the observational distribution.
- The source has the smallest variance; repeatedly comparing conditional variances yields the causal order.
- After the order, regression or sparse regression estimates the parents and edge coefficients.
- Check variable scales, measurement error, hidden confounding and violations of the equal-variance assumption.
- The complete DAG is obtained not from Gaussianity alone but from the additional assumption of a common structural error variance.

8 Selected exercises

Motivation. The exercises give a bottom-up version of the variance ordering based on the precision matrix, show concretely how a change of units can reverse the inferred direction, and prove the conditional-variance decomposition behind the US algorithm.

Exercise 13.1 (Bottom-up ordering from the precision matrix)

In an equal-variance Gaussian SEM with nonzero edge coefficients, let $\Omega = \Sigma^{-1}$.
(a) Show that $\Omega_{jj} = \{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}/\sigma^2$, where $\mathrm{ch}(j)$ is the set of children of j. Deduce that $\mathrm{Var}(X_j \mid X_{-j}) = \sigma^2$ if j is a sink (has no children) and $< \sigma^2$ otherwise.
(b) Show that after deleting a sink, the remaining variables again follow an equal-variance Gaussian SEM on the induced subgraph, so that repeatedly removing $\arg\min_j \Omega_{jj}$ yields a reversed causal order.

Solution [Supplement].

(a)

Step 1. As in Ch. 9, $\Omega = (I - B)^\top (I - B)/\sigma^2$, so $\Omega_{jj} = \sum_l (I - B)_{lj}^2/\sigma^2$. The term $l = j$ is 1 (no self-loops), and for $l \neq j$, $(I - B)_{lj} = -b_{lj}$, which is nonzero exactly when $j \to l$.

Step 2. By the node-wise theorem of Ch. 8, $\mathrm{Var}(X_j \mid X_{-j}) = 1/\Omega_{jj} = \sigma^2/\{1 + \sum_{l \in \mathrm{ch}(j)} b_{lj}^2\}$, which equals σ^2 iff $\mathrm{ch}(j) = \varnothing$.

(b)

Step 3. A sink j is nobody's parent, so it does not appear on the right side of any other structural equation. The equations of the other variables, with their independent $N(0, \sigma^2)$ errors, form an equal-variance Gaussian SEM on the DAG with j removed, and they determine the joint law of X_{-j}.

Step 4. In the current set, every sink has $\Omega_{jj} = 1/\sigma^2$ and every non-sink has $\Omega_{jj} > 1/\sigma^2$, so the argmin is a sink. Removing it and repeating lists the variables so that each one has no children among the variables still remaining, i.e. in reverse causal order. ■

Exercise 13.2 (A change of units can reverse the direction)

Let $X = \epsilon_X$, $Y = bX + \epsilon_Y$, $\epsilon_X, \epsilon_Y \overset{\text{ind}}{\sim} N(0, \sigma^2)$, $b \neq 0$, and $\tilde{Y} = sY$ with $s > 0$.
(a) Show that $(X, \tilde{Y})$ follows the Gaussian SEM $X \to \tilde{Y}$ with error variances σ^2 and $s^2\sigma^2$, which is equal-variance only for $s = 1$.
(b) Show that the top-down rule applied to $(X, \tilde{Y})$ selects $\tilde{Y}$ first iff $s^2 < 1/(1 + b^2)$.
(c) Show that for $s = 1/(1 + b^2)$ the pair $(X, \tilde{Y})$ is *exactly* an equal-variance Gaussian SEM with $\tilde{Y} \to X$.

Solution [Supplement].

(a) $\tilde{Y} = sbX + s\epsilon_Y$ with $s\epsilon_Y \sim N(0, s^2\sigma^2)$ independent of X.

(b) $\mathrm{Var}X = \sigma^2$ and $\mathrm{Var}\tilde{Y} = s^2(1 + b^2)\sigma^2$; the rule picks the smaller variance first, which is $\tilde{Y}$ iff $s^2(1 + b^2) < 1$.

(c)

Step 1. By the two-variable computation of this chapter, the reverse factorization of (X, Y) is $Y = \epsilon'_Y$, $X = cY + \eta$ with $\mathrm{Var}\eta = \sigma^2/(1 + b^2)$ and η independent of Y. Since $\tilde{Y}$ is a rescaling of Y, $X = (c/s)\tilde{Y} + \eta$ with the same η, independent of $\tilde{Y}$.

Step 2. The errors of the reverse SEM have variances $\mathrm{Var}\tilde{Y} = s^2(1 + b^2)\sigma^2$ and $\sigma^2/(1 + b^2)$. For

$s = 1/(1+b^2)$ both equal $\sigma^2/(1+b^2)$, and $(X, \tilde{Y})$ is Gaussian, so this is an equal-variance Gaussian SEM with $\tilde{Y} \to X$.

The method then outputs the wrong direction with full confidence; nothing in the data reveals the error. Whether the equal-variance assumption holds depends on the chosen units, which is the scale dependence warned about above. ■

Exercise 13.3 (Conditional variances in an additive noise model)

In an ANM $X_j = f_j(X_{\mathrm{pa}(j)}) + \epsilon_j$ with independent errors, $\mathbb{E}\epsilon_j = 0$, $\mathrm{Var}\epsilon_j = \sigma_j^2$ and $\mathbb{E}f_j(X_{\mathrm{pa}(j)})^2 < \infty$, let S be an ancestral set and $j \notin S$.
(a) Show that ϵ_j is independent of $(X_S, X_{\mathrm{pa}(j)})$.
(b) Show that $\mathrm{Var}(X_j \mid X_S) = \sigma_j^2 + \mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}$ almost surely; hence $U_j(S) = \sigma_j^2 + \mathbb{E}[\mathrm{Var}\{f_j(X_{\mathrm{pa}(j)}) \mid X_S\}]$, which equals σ_j^2 when $\mathrm{pa}(j) \subseteq S$.

Solution [Supplement].

(a)

Step 1. By induction along a causal order, each X_k is a measurable function of $(\epsilon_l)_{l \in \mathrm{an}(k) \cup \{k\}}$.

Step 2. For $k \in S$, $\mathrm{an}(k) \cup \{k\} \subseteq S$ because S is ancestral, and $j \notin S$. For $k \in \mathrm{pa}(j)$, $\mathrm{an}(k) \cup \{k\} \subseteq \mathrm{an}(j)$, which does not contain j by acyclicity. So $(X_S, X_{\mathrm{pa}(j)})$ is a function of $(\epsilon_l)_{l \neq j}$, which is independent of ϵ_j.

(b)

Step 3. Write $F = f_j(X_{\mathrm{pa}(j)})$ and $\bar{F} = \mathbb{E}(F \mid X_S)$. By (a), $\mathbb{E}(X_j \mid X_S) = \bar{F} + \mathbb{E}\epsilon_j = \bar{F}$, so

$$\mathrm{Var}(X_j \mid X_S) = \mathbb{E}\{(F - \bar{F})^2 \mid X_S\} + 2\,\mathbb{E}\{(F - \bar{F})\epsilon_j \mid X_S\} + \mathbb{E}(\epsilon_j^2 \mid X_S).$$

Step 4. Since ϵ_j is independent of (F, X_S), $\mathbb{E}\{(F - \bar{F})\epsilon_j \mid X_S\} = \mathbb{E}\epsilon_j \cdot \mathbb{E}(F - \bar{F} \mid X_S) = 0$ and $\mathbb{E}(\epsilon_j^2 \mid X_S) = \sigma_j^2$. The first term is $\mathrm{Var}(F \mid X_S)$. Taking expectations gives $U_j(S)$.

Step 5. If $\mathrm{pa}(j) \subseteq S$, then F is a function of X_S, so $\mathrm{Var}(F \mid X_S) = 0$ and $U_j(S) = \sigma_j^2$. ■